

Evleen Al-Dogom

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PROFESSIONAL SUMMARY

Motivated Computer Science with Artificial Intelligence student at Royal Holloway, University of London, with practical research internship experience in quantitative finance software engineering and AI-driven applications. Demonstrated ability to design and implement options pricing models, volatility calibration systems, and fullstack web applications across Python, Java, and React. Passionate about applying artificial intelligence and quantitative methods to real-world financial systems, with a long-term ambition to build a career as an AI Engineer in the FinTech or quantitative finance sector.

EDUCATION

Royal Holloway, University of London | Egham, United Kingdom

BSc Computer Science with Artificial Intelligence Expected Graduation: June 2028

Relevant Modules: Software Engineering · Algorithms & Complexity · Artificial Intelligence · Symbolic AI · Operating Systems · Information Security · Database Systems · Multidimensional Data Processing · ObjectOriented Programming (Java) · C Programming · SQL · Data Structures

TECHNICAL SKILLS

- **Programming Languages:** Java · Python · SQL · C · JavaScript
- **Web Development:** React · HTML · CSS · REST APIs · Shopify Hydrogen
- **Backend & Frameworks:** Spring Boot · Maven · JDBC · JWT Authentication
- **Databases:** PostgreSQL · MariaDB · H2 Database
- **AI & Data:** Symbolic AI · Machine Learning Fundamentals · Numerical Optimisation · Data Processing
- **Quantitative Finance:** SABR Volatility Model · Options Pricing · Implied Volatility · Volatility Smile Calibration · Financial Modelling
- **Developer Tools:** Git · GitLab · IntelliJ IDEA · Eclipse · JaCoCo · Checkstyle · SpotBugs
- **Languages:** English (Fluent) · Arabic (Native)

EXPERIENCE

Research Software Engineering Intern

Professor's Research Group - Royal Holloway, University of London

- Implemented the SABR (Stochastic Alpha Beta Rho) stochastic volatility model in Python, enabling volatility smile calibration across multiple strike prices and maturities for options pricing research.
- Developed an RMSE-based optimisation pipeline to calibrate SABR model parameters, improving the accuracy of the implied volatility surface fit against live market data.
- Applied Hagan's SABR approximation formula to compute analytical implied volatilities, translating complex mathematical derivations into production-quality numerical code.
- Collaborated with a university researcher to debug, refactor, and extend an existing quantitative finance codebase, improving maintainability and numerical accuracy.

PROJECTS

Options Pricing & Volatility Calibration Tool

Python · React · Shopify Hydrogen · SABR Model · Numerical Optimisation

- Designed and built a full-stack options pricing application integrating the SABR stochastic volatility model to compute and visualise implied volatility surfaces.
- Implemented volatility smile fitting via RMSE-based calibration, optimising model parameters to match real-world market data with high precision.
- Integrated a Shopify Hydrogen (headless React) frontend with a Python backend to deliver a browseraccessible, interactive research tool.
- Applied quantitative finance concepts - financial derivatives, implied volatility, and stochastic volatility modelling - to produce a research-grade application.

Restaurant Management System (ARMS)

Java · Spring Boot · Maven · REST APIs · MariaDB · JWT Authentication · Flyway · Git

- Led full-stack development of a multi-role restaurant management system within a software engineering team, delivering customer ordering, waiter confirmation workflows, and a manager dashboard.
- Implemented JWT-based authentication and role-based access control to secure API endpoints across customer, waiter, and manager roles.
- Integrated Stripe payment processing into the backend, enabling end-to-end transaction handling within the live application.
- Achieved 90%+ unit and integration test coverage using JaCoCo and enforced code quality with Checkstyle and SpotBugs across the full codebase.
- Rebuilt the complete React/Vite frontend independently after the original subgroup failed to deliver, ensuring on-time project completion.
- Managed GitLab workflows with issue-tracked branches, per-file commits, and detailed merge request documentation, maintaining a clean and auditable development history.

Relational Database Design - Database Systems Coursework

PostgreSQL · SQL · JDBC · Entity-Relationship Modelling · Normalisation

- Designed a fully normalised relational database schema from business requirements, applying 1NF–3NF principles to eliminate redundancy.
- Implemented and queried the database using PostgreSQL and JDBC, demonstrating proficiency in constraint design, relational modelling, and SQL querying.